A C PROGRAM FUNCTION THAT CONSTRUCTS A MIN HEAP FROM AN ARRAY OF ELEMENTS.

```
#include <stdio.h>
void minHeapify(int arr[], int n, int i)
{
    int smallest = i;  // Initialize smallest as root
    int left = 2 * i + 1;  // left child
    int right = 2 * i + 2; // right child
    if (left < n && arr[left] < arr[smallest])
        smallest = left;
    if (right < n && arr[right] < arr[smallest])
        smallest = right;
    if (smallest != i)
 {
         int temp = arr[i];
        arr[i] = arr[smallest];
        arr[smallest] = temp;
        minHeapify(arr, n, smallest);
    }
 }
 void buildMinHeap(int arr[], int n)
  {
  for (int i = n / 2 - 1; i >= 0; i--)
        minHeapify(arr, n, i);
 }
 void printArray(int arr[], int n)
  {
    for (int i = 0; i < n; ++i)
        printf("%d ", arr[i]);
    printf("\n");
 }
 int main()
 {
    // Test with a random array
    int randomArray[] = {4, 10, 3, 5, 1};
    int n1 = sizeof(randomArray) / sizeof(randomArray[0]);
    printf("Original Random Array: ");
    printArray(randomArray, n1);
    buildMinHeap(randomArray, n1);
    printf("Min Heap from Random Array: ");
    printArray(randomArray, n1);
    printf("\n");
    int sortedArray[] = {8, 7, 5, 4, 2};
    int n2 = sizeof(sortedArray) / sizeof(sortedArray[0]);
    printf("Original Sorted Array: ");
    printArray(sortedArray, n2);
    buildMinHeap(sortedArray, n2);
    printf("Min Heap from Sorted Array: ");
    printArray(sortedArray, n2);
    return 0;
 }
```